

DOCUMENT REVISION HISTORY

Version	Date	Author	Description of Changes
v1.0	10.06.2026	Cognitive Crew	Initial draft - SRS skeleton completed per IEEE Std 830-1998 template; project header (Section 1), team details, and overall document structure established.
v1.1	12.06.2026	Farah Hamna M	Authored Section 1 (Introduction): Purpose, Scope, and the Definitions/Acronyms/Abbreviations table for the AlertGov platform.
v1.2	15.06.2026	Rishikesh V	Drafted Section 2.1-2.2 (Product Perspective and Product Functions - Feature Summary) describing the 12 core platform functions from incident reporting to multi-tier broadcast.
v1.3	17.06.2026	Shruthin K	Completed Section 2.3 (User Classes and Characteristics), defining the 5-tier hierarchy: Village EOC, Tahsildar, District EOC, District Collector, and State SDMA Admin.
v1.4	19.06.2026	Farah Hamna M	Documented Section 3.1-3.2 (System Architecture and Microservices Breakdown) detailing 11 Spring Boot microservices, Eureka discovery, and Spring Cloud Gateway.
v1.5	22.06.2026	Rishikesh V	Added Section 3.3 (Technology Stack), confirming React 19, Vite, Leaflet GIS, Java 17 Spring Boot 3.2.4, Oracle XE, MongoDB, and JWT bearer authentication.
v1.6	24.06.2026	Shruthin K	Documented Section 3.5 (Data Flow Description), tracing the end-to-end incident submission, verification queue, escalation, collector approval, and multi-channel broadcast across 8 steps.
v1.7	27.06.2026	Farah Hamna M	Drafted Section 4.1 (User Interfaces) for Village Field Reporting, Taluk Incident Verification Queue, and District Emergency Command dashboards.
v1.8	30.06.2026	Rishikesh V	Extended Section 4.1 to cover District Collector Approval, State SDMA 38-District Monitor, and Analytics dashboards, including page routes and backing REST endpoints.
v1.9	03.07.2026	Shruthin K	Completed Section 4.3-4.4 (Software and Communication Interfaces), specifying REST/JSON microservice communication, WebSocket/SSE real-time events, and Open-Meteo weather APIs.
v1.10	07.07.2026	Farah Hamna M	Authored Section 5.1-5.2 functional requirements for the Authentication & RBAC Service and the Incident Lifecycle Management Service (FR-AUTH, FR-INC).
v1.11	10.07.2026	Rishikesh V	Authored Section 5.3-5.4 functional requirements for the Multi-Tier Approval Workflow Service and Public Alert Broadcast Service (FR-APP, FR-ALT).
v1.12	14.07.2026	Shruthin K	Authored Section 5.5-5.8 functional requirements covering Notification Dispatch, Executive Analytics, Weather Advisory, and GIS Geo-Fencing Services.
v1.13	17.07.2026	Farah Hamna M	Documented Section 5.9-5.10 (Bilingual Localization & Audit Logging) and mapped role permissions across all 38 revenue districts of Tamil Nadu.
v1.14	21.07.2026	Rishikesh V	Drafted Section 6.1 (Local LLM Disaster Risk & Bilingual Instruction Generator), specifying Ollama / Llama 3 prompt engineering, risk scoring, and English/Tamil formatting.
v1.15	24.07.2026	Shruthin K	Drafted Section 6.2 (Live Weather Advisory & CircuitBreaker), integrating Open-Meteo API with Resilience4j fallback mechanisms during external network outages.
v1.16	28.07.2026	Farah Hamna M	Authored Section 7 (Non-Functional Requirements), specifying <2s API latency, 99.9% uptime, AES-256 encryption, RBAC security, and NDMA/TNDMA compliance targets.
v1.17	31.07.2026	Rishikesh V	Authored Section 8.1-8.2 (Database Design & ER Overview), detailing the dual-database model with 10 Oracle XE relational tables and MongoDB multimedia collections.
v1.18	04.08.2026	Farah Hamna M	Added database entity dictionaries for ALERTS, INCIDENTS, APPROVAL_WORKFLOWS, USERS, and AI_PREDICTIONS into Section 8.
v1.19	07.08.2026	Rishikesh V	Authored Section 9 (API Design Overview), documenting all 71 confirmed REST endpoints across the 11 microservice capability areas.
v1.20	11.08.2026	Shruthin K	Drafted Section 10 (DevOps & Deployment Plan with Jenkins CI/CD & Docker Compose) and Section 11 (Project Timeline & Milestones across 8 sprint weeks).
v1.21	14.08.2026	Farah Hamna M	Completed Section 12 (Team Composition & Role Assignment) and Section 13 (Risk Register with 10 project-specific risks and mitigations).
v1.22	18.08.2026	Shruthin K	Compiled Section 14 (Appendix): Glossary of Domain-Specific Terms, API Request-Response Samples, and IEEE/TNDMA References & Citations.
v1.23	25.08.2026	Rishikesh V	Incorporated mentor feedback; standardized 5-tier administrative workflow nomenclature and verified Tamil Unicode rendering consistency across all modules.
v1.24	02.09.2026	Shruthin K	Added system architecture diagrams, verification flowcharts, and GIS interactive mapping references to support Section 4 and Section 5 functional specs.
v2.0	15.09.2026	Farah Hamna M	Post Internal Review - Consolidated formatting across all 30 pages, validated all 71 REST API contracts, and finalized the complete IEEE Std 830-1998 SRS for official submission.